Making Solutions and Dilutions

Chem 151 Lab

Today's Lab

- Gain skills in calculating and making solutions with particular molarities (M; mol/L)
- We will then use these solutions next week to experimentally determine the equilibrium constant (K) for the following reaction:

$$Fe^{3+}(aq) + SCN^{-}(aq) \rightleftharpoons FeSCN^{2+}(aq)$$

$$K_c = \frac{[FeSCN^{2+}]_{eq}}{[Fe^{3+}]_{eq}[SCN^{-}]_{eq}}$$

Making 3 Stock Solutions

1. Make 100 mL of an aqueous 0.1 M Fe^{3+} solution from $Fe(NO_3)_3 \bullet 9H_2O(s)$
   - Use the pre-mixed **nitric acid solution** to dilute the $Fe(NO_3)_3 \bullet 9H_2O$.

2. Make 100 mL of an aqueous 20 mM Fe^{3+} solution from $Fe(NO_3)_3 \bullet 9H_2O(s)$
   - Use the pre-mixed **nitric acid solution** to dilute the $Fe(NO_3)_3 \bullet 9H_2O$.

3. Make 100 mL of an aqueous 20 mM SCN^- solution from KSCN(s)
   - Use the pre-mixed **nitric acid solution** to dilute the KSCN.

Making a Stock Solution Calculation Example

- Make 100 mL of an aqueous 20 mM Fe^{3+} solution from $Fe(NO_3)_3 \bullet 9\ H_2O(s)$

MW $Fe(NO_3)_3 \bullet 9\ H_2O(s)$ = 403.999 g/mol

$$20\ mM\left(\frac{1M}{1000\ mM}\right) = 0.020\ M = 0.020\ \frac{mol}{L}$$

100 mL = 0.100 L

$$0.100\ L\left(\frac{0.020\ mol\ Fe^{3+}}{1\ L}\right)\left(\frac{1\ mol\ Fe(NO_3)_3 \bullet 9\ H_2O}{1\ mol\ Fe^{3+}}\right)\left(\frac{403.999\ g}{1\ mol\ Fe(NO_3)_3 \bullet 9\ H_2O}\right) = 0.808\ g\ Fe(NO_3)_3 \bullet 9\ H_2O$$

How to Make that Solution in the Lab:

$$0.100\ L\left(\frac{0.020\ mol\ Fe^{3+}}{1\ L}\right)\left(\frac{1\ mol\ Fe(NO_3)_3{\bullet}9\ H_2O}{1\ mol\ Fe^{3+}}\right)\left(\frac{403.999\ g}{1\ mol\ Fe(NO_3)_3{\bullet}9\ H_2O}\right) = 0.808\ g\ Fe(NO_3)_3{\bullet}9\ H_2O$$

- Weigh out 0.81 g of $Fe(NO_3)_3{\bullet}9\ H_2O(s)$ and transfer solid to 100.0 mL volumetric flask
- Add 0.5 M nitric acid in dH_2O to fill approximately ½ the volumetric
- Swirl to mix until fully (or mostly) dissolved
- Add the premixed 0.5 M nitric acid solution carefully until the bottom of the meniscus is at the 100.0 mL mark (do not overfill!)
- Cover the top of the volumetric and mix thoroughly by inverting at least 50 times

You just made 100 mL of a 20 mM Fe^{3+} stock solution! Label and store.

Using Stock Solutions to Make New, Diluted Solutions (you will be making 4 of these):

Dilution!

$M_1V_1 = M_2V_2$ OR $C_1V_1 = C_2V_2$

Example: From your stock 20 mM $Fe^{3+}(aq)$ solution, make 50 mL of an aqueous 8 mM $Fe^{3+}(aq)$ solution

$$V_2 = \frac{M_1V_1}{M_2} = \frac{8\ mM \times 50\ mL}{20\ mM} = 20\ mL$$

What does this mean?

You would use a volumetric pipette to pull 20.0 mL of 20 mM Fe^{3+} stock and transfer that into a 50 mL volumetric flask. Then you would add 30.0 mL of the stock nitric solution to get a total volume of 50.0 mL of 8 mM Fe^{3+}

Soln #	0.100 M Fe^{3+}	0.200 mM SCN^-	0.5 M HNO_3	Final Volume	$[SCN^-]_{initial}$ (in M)	$[Fe(SCN)]^{2+}_{eq}$ (in M)
Blank	10.0 mL	0.0 mL	20.0 mL	**30.0 mL**	_____	_____
1	10.0 mL	5.0 mL	15.0 mL	**30.0 mL**	_____	_____
2	10.0 mL	10.0 mL	10.0 mL	**30.0 mL**	_____	_____
3	10.0 mL	20.0 mL	0.0 mL	**30.0 mL**	_____	_____

$[FeSCN^{2+}]_{eq} = [SCN^-]_{initial}$ for next week's lab, so

Dilution!

$M_1V_1 = M_2V_2$

Solve for M_2:

Soln #	Fe^{3+}	$[Fe^{3+}]_{initial}$	0.2 mM SCN^-	$[SCN^-]_{initial}$
4	10.0 mL of 4.00 mM	_____	10.0 mL	_____
5	10.0 mL of 6.00 mM	_____	10.0 mL	_____
6	10.0 mL of 8.00 mM	_____	10.0 mL	_____

Dilution!

$M_1V_1 = M_2V_2$

Solve for M_2:

Note: Total volume is always 20 mL here!

Logistics for Today's Lab

You should have 7 solutions properly labeled and stored in your lab drawer for next weeks lab

3 Stock Solutions

4 Dilution Solutions

*Make sure you dispose of all waste into the proper waste containers in the hood

*Be extra careful not to cross-contaminate any stock bottles today!!